

Linked Lists implementation using C

Topic 3

Creation of Linked List Node

Search for a specific element in the linked list

Insert a node at a given position in the linked list

Delete a node at a given position in the linked list

Learning Objectives

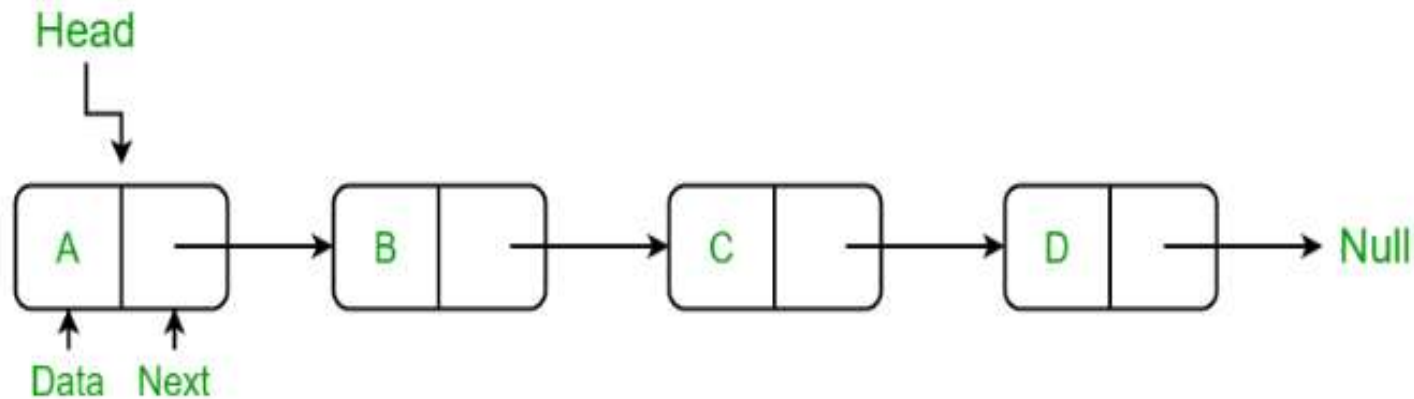
By the end of this chapter, you will be able to:

- Discuss the Creation of Linked List Node
- Explain the Concepts of Searching for a specific element in linked list
- Explain the Concepts of Inserting and deleting a node in the linked list



Linked List

A linked list is a linear data structure used to store and organize a collection of elements (nodes) in a sequential manner. Each node in a linked list contains two components: data and a reference (pointer) to the next node in the sequence. The data component holds the actual value or information of the element, while the next component points to the next node in the list.



Creation of Linked List Node

```
//Linear singly listed list CREATENODE
```

```
#include<stdio.h>
```

```
#include <conio. h>
```

```
#include<alloc.h>
```

```
struct node
```

```
{
```

```
    int info;
```

```
    struct node *next;
```

```
};
```

```
struct node*head=NULL, *tail;
```

```
struct node *createnode()
```

```
{
```

```
    struct node *temp;
```

```
    temp=(struct node *)malloc(sizeof(struct node));
```

```
    printf("Enter the data....");
```

```
    scanf("%d",&temp->info);
```

```
    temp->next=NULL;
```

```
    return temp;
```

```
}
```

Creation of Linked List Node

```
void display() {  
    struct node *t;  
    for(t=head;t!=NULL;t=t-> next) {  
        printf("%d=",t->info);  
    }  
}
```

```
void main() {  
    char ans='y';  
    struct node *newnode;  
    clrscr();
```

```
    while(ans=='y' || ans=='Y'){  
        newnode =createnode;  
        if(head==NULL){  
            head=newnode;  
            tail=newnode;  
        }  
        else{  
            tail->next=newnode;  
            tail=newnode;  
        }  
        fflush();  
        printf(" want to continue...(y/n)");  
        scanf("%c",&ans ) ;  
    }  
    printf("\n");  
    printf("Elements are");  
    display;  
}
```

Search for a specific element in the linked list

// Function to search for a specific element in the linked list

```
struct Node* searchNode(struct Node* head, int data) {  
    struct Node* temp = head;  
    while (temp != NULL) {  
        if (temp->data == data) {  
            return temp;  
        }  
        temp = temp->next;  
    }  
    return NULL;  
}
```

Insert a node at a given position in the linked list

// Function to insert a node at a given position in the linked list

```
void insertAtPosition(struct Node** head, int data, int position) {  
    if (position <= 0) {  
        insertAtBeginning(head, data);  
    }  
    else {  
        struct Node* newNode = createNode(data);  
        struct Node* temp = *head;  
        for (int i = 1; i < position; i++) {
```

```
            if (temp == NULL) {  
                printf("Position exceeds the length of the  
                    linked list. Inserting at the end.\n");  
                insertAtEnd(head, data);  
                return;  
            }  
            temp = temp->next;  
        }  
        newNode->next = temp->next;  
        temp->next = newNode;  
    }  
}
```

Delete a node at a given position in the linked list

```
// Function to delete a node with the given data  
from the linked list
```

```
void deleteNode(struct Node** head, int data) {  
    if (*head == NULL) {  
        printf("Linked list is empty. Cannot delete.\n");  
        return;  
    }  
    if ((*head)->data == data) {  
        struct Node* temp = *head;  
        *head = (*head)->next;  
        free(temp);  
        return;  
    }  
}
```

```
    struct Node* temp = *head;  
    while (temp->next != NULL) {  
        if (temp->next->data == data) {  
            struct Node* toDelete = temp->next;  
            temp->next = temp->next->next;  
            free(toDelete);  
            return;  
        }  
        temp = temp->next;  
    }  
    printf("Data not found in the linked list. Cannot  
delete.\n");  
}
```

THANK YOU